

Unit – III : Magnetic Materials

Introduction :

Magnetic Materials are the materials which can be made to behave as magnets. When these materials are kept in an external magnetic field, they will create a permanent magnetic moment in it.

Magnetism originates from the magnetic moment of magnetic materials due to the rotational motion of the charged particles. When an electron revolves around the positive nucleus orbital magnetic moment arises and due to the spinning of electrons, spin magnetic moment arises.

Basic Definitions:

1. Magnetic dipole moment :

A system having two opposite magnetic poles separated by a distance 'd' is called as a magnetic dipole. If 'm' is magnetic pole strength and 'l' is the length of the magnet, then its dipole moment is given by

$$\mu_m = m.l$$

Magnetic moment can also be defined as $\mu_m = i.a$, where 'i' is the electric current that flows through a circular wire of an area of cross section 'a'.

2. Bohr Magnetron :

The orbital magnetic moment and the spin magnetic moment of an electron in an atom can be expressed in terms of atomic unit of magnetic moment called bohr magnetic

$$1 \text{ Bohr Magnetron} = \frac{e\hbar}{2m} = \mu_B = 9.27 \times 10^{-24} \text{ AM}^2$$

3. Magnetic field

The space around the magnet or the current carrying conductor where the magnetic effect is felt is called Magnetic field.

4. Magnetic lines of force

A Magnetic line of force is defined as the continuous curve in a magnetic field. The tangent drawn at any point on the curve gives the direction of the resultant Magnetic intensity at that point.

5. Magnetic induction or Magnetic flux density (B)

It is defined as the number of Magnetic lines of force passing normally through unit area of cross section 'A' at that point

$$\vec{B} = \frac{\vec{\Phi}_m}{A} \text{ Wb/m}^2 \quad \text{where } \vec{B} = B \cos \theta$$

6. Magnetic field intensity (H)

It is defined as the force experienced by an unit north pole placed at the given point in a Magnetic field.

$$\vec{H} = \frac{\vec{F}}{m} \text{ Wb/m}^2$$

$$(i.e) \quad \vec{H} = \frac{m}{4\pi\mu_0\mu_r r^2} \cdot \vec{r} \text{ N/Wb or Ampere/meter}$$

7. Magnetisation (or) Intensity of Magnetisation (I)

The term magnetization is the process of converting a non magnetic material into a magnetic material. It measures the magnetization of the magnetized specimen.

It is also defined as the magnetic moment per unit volume.

$$I = \frac{M}{V} = \frac{m \cdot l}{l \cdot a} = \frac{m}{a} \text{ weber/m}^2$$

8. Magnetic Susceptibility (χ_m)

It is defined as the ratio between the magnetic flux density (B) and the magnetic field intensity(H).

$$\chi_m = \frac{I}{H}$$

9. Magnetic Permeability (μ)

It is defined as the ratio between the magnetic flux density (B) and the magnetic field intensity (H).

$$\mu = \mu_0 \mu_r = \frac{B}{H}$$

10. Retentivity (or) Remanence

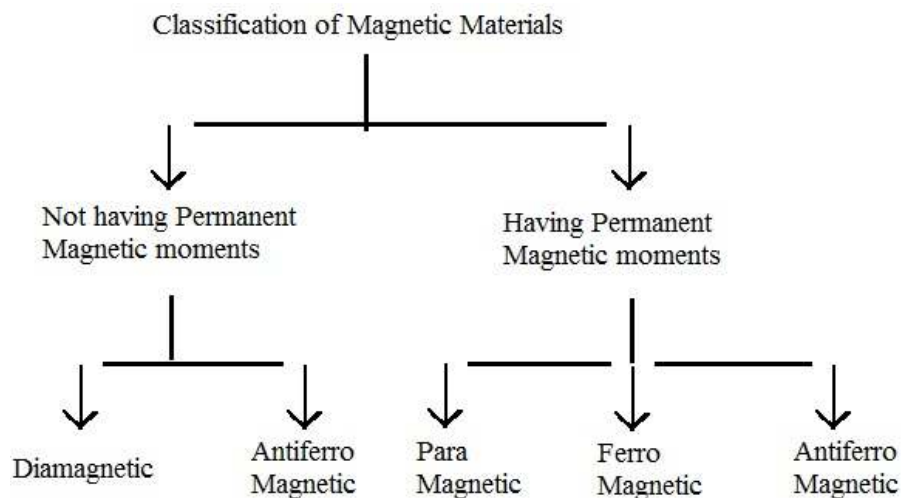
When the external magnetic field is applied to a magnetic material is removed, the magnetic material will not loss its magnetic property immediately there exists some residual intensity of magnetization in the specimen even when the magnetic field is cut off. This is called residual magnetism or retentivity.

11. Coactivity

The residual magnetism can be completely removed from the material by applying a reverse magnetic field. Hence coercivity of the magnetic material is the strength of reverse magnetic field ($-H_c$) which is used to completely demagnetize the material.

Classification of magnetic materials on the casis of magnetic moment

Magnetic materials are classified according to the presence or absence of the permanent magnetic dipoles.

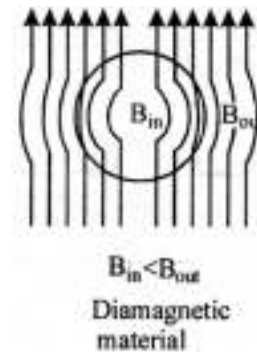


Diamagnetic material

In a diamagnetic material, the electron orbits are more or less random, and mostly all the magnetic moments are cancelled. The spins are almost paired, they have even number of electrons and has equal number of electrons spinning in two opposite directions. Hence the net magnetic moment is zero.

Effect of external field:

When an external magnetic field is applied the electrons reorient in such a way that they align perpendicular to the field direction and their magnetic moments oppose the external magnetic field. This will reduce the magnetic induction present in the specimen.



Properties:

1. They repel the magnetic lines of force
2. Susceptibility is negative and it is independent of temperature and applied magnetic field strength.
3. Permeability is less than 1.
4. There is no permanent dipole moment so they are called weak magnets.
5. When temperature is less than critical temperature diamagnetic become normal material

Eg: Gold, Germanium, Silicon etc.

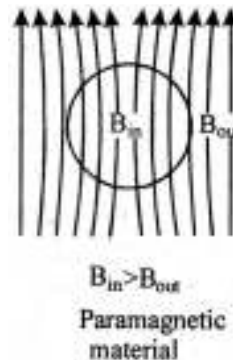
Paramagnetic materials.

In the case of paramagnetic materials the spins are in two opposite directions and they will not be equal. There exists some unpaired electrons which give rise to spin magnetic moment. Hence the resultant magnetic moment will not be equal to zero.

However the absence of external field is the magnetic moments are oriented randomly. Due to its random orientation some magnetic moment gets cancelled and the material possesses very less magnetization in it.

Effect of external field.

When an external field is applied, the magnetic moments of individual molecules reorient itself along the direction of the magnetic field and the material is magnetized.



Properties

1. The magnetic lines of force pass through the material.
2. Magnetic susceptibility is positive and it is given by

$$\chi = \frac{C}{T - \theta} \text{ (Curie-Weiss law)}$$

Where C – Curie constant

T – Absolute temperature

Q – Curie temperature

3. Permeability is greater than one.
4. They possess permanent dipole moment
5. When the temperature is less than curie temperature paramagnetic materials becomes diamagnetic material

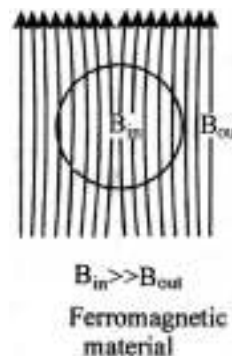
Eg: CuSO_4 , MnSO_4 , & Platinum etc.

Ferromagnetism.

In a Ferromagnetic material the number of unpaired electrons is more. Most of these spin magnetic moments point in one direction. Hence even in the absence of external field, the magnetic moments align themselves parallel to each other and give rise to magnetic field.

Effect of Magnetic field.

To these materials even if a small external magnetic field is applied, the magnetic moments which are already aligned parallel, reorient itself along the direction of the magnetic field and they become very strong magnets.



Properties.

1. Since some magnetization is already existing in these materials, all the magnetic moments lines of force pass through it.
2. They have permanent dipole moment. So they act as strong magnets.
3. They exhibit magnetization even in the absence of external field. This property is called Spontaneous magnetization.
4. Its Susceptibility is positive and high and it is given by

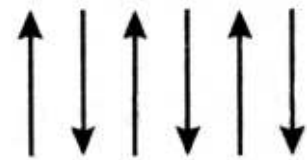
$$\chi_m = \frac{C}{T - \theta}$$

5. When the temperature is greater than Curie temperature, ferromagnetic material becomes paramagnetic material.
6. Permeability is very much greater than 1.

Eg: Ni, Co, Fe, Gd, etc.

Anti – Ferromagnetism

In this the spins are aligned in anti parallel manner due to unfavorable exchange interaction among them, resulting in zero magnetic moment. Even when the field is increased, it has almost zero induced magnetic moment.



Properties.

1. The Susceptibility is very small and is positive. It is given by

$$\chi_m = \frac{C}{T + \theta} \quad \text{for} \quad T > T_N$$

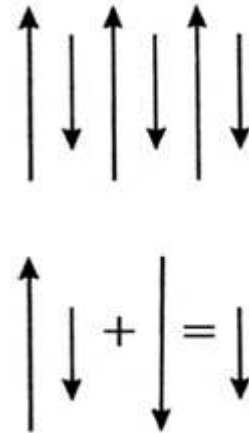
T_N is the Neel temperature

2. Initially, the susceptibility increases slightly as the temperature increases and beyond a particular temperature known as Neel temperature, the susceptibility decreases with temperature.

Eg: Ferrous Oxide, MnS, FeCl₂ etc.

Ferri Magnetism.

It is a special case of magnetic material and it is composed of two sets of different transition metal ions having different values of magnetic moment with anti parallel alignment as shown below.



These materials have anti parallel magnetic moments of different magnitudes giving rise to fairly large magnetic moment in the presence of external magnetic field.

Properties.

1. The Susceptibility is Very large and is positive represented by

$$\chi_m = \frac{C}{T - \theta} \quad \text{when } T > T_N$$

2. Beyond the Neel temperature χ decreases
3. These materials have low eddy current losses and low hysteresis losses.

Quantum Theory of Paramagnetism

According to classical mechanics, the magnetic moments of individual atoms (ions) behave as tiny bar magnets capable of orienting themselves in any arbitrary directions with the applied magnetic field. But this cannot be true. According to quantum mechanics the magnetic moment of an atom or an ion is not a freely rotating one but it is restricted to a finite set of orientations relative to the applied field.

In the absence of the field, the total angular momentum quantum number J is obtained by the sum of the total spin and the orbital angular momentum quantum number using $L - S$ coupling. In the presence of the field, there are $(2J+1)$ possible orientations of J with the applied magnetic field.

The magnetic moment of an atom or an ion $\mu = g \mu_B J$ (1)

Where $\mu_B = \frac{e\hbar}{2m} = 9.27 \times 10^{-24} \text{ Am}^2$ is known as Bohr magneton J is the total angular momentum and 'g' the Lande g – factor or spectroscopic splitting factor. The

total angular momentum J is given by the sum of the orbital angular momentum and spin angular momentum. The g – factor is given by

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \quad (2)$$

According to quantum mechanics, since there are (2J+1) possible combinations of orientations.

Therefore Eqn. (1) can be written as $\mu = m_J g \mu_B$ (3)

Where m_J is the azimuthal quantum number and it has (2J+1) possible values such as J, (J-1), (J-2).....,0,..... (J-2), -(J-1), - J. The energy level of a system in magnetic field is given as

$$U = - \mu \mu_o H \quad (4)$$

Sub eqn.(3) in eqn.(4)

$$U = - g m_J \mu_B \mu_o H \quad (5)$$

Substituting different Values of m_J in eqn. (5) the corresponding (2J+1) Values of energy in the magnetic field H are written as.

$$U = g \mu_B J \mu_o H, g \mu_B (J-1) \mu_o H, \dots \dots \dots - g \mu_B (J-1) \mu_o H, -g \mu_B J \mu_o H \quad (6)$$

The average magnetic moment of the system can be obtained Using Maxwell–Boltzmann distribution function.

$$\bar{\mu} = \frac{\text{Total magnetic moment of all dipoles}}{\text{number of dipoles per unit volume}}$$

From Langavenin's theory of paramagnetism the number of dipoles with maximum energy (cos θ) per unit volume can be written as

$$n = n_o \exp \left[\frac{-U}{k_B T} \right] \quad (7)$$

i.e $\bar{\mu} = \frac{\mu n}{n}$ (8)

Sub eqn. (7) and (8)

$$\bar{\mu} = \frac{\mu n_o \exp \frac{-U}{k_B T}}{n_o \exp \left[\frac{-U}{k_B T} \right]} \quad (9)$$

Sub eqn. (3) and eqn. (5) in eqn. (9)

$$\bar{\mu} = \frac{m_J g \mu_B \exp\left(\frac{-(-g \mu_B m_J \mu_o H)}{K_B T}\right)}{\exp\left(\frac{-(-g \mu_B m_J \mu_o H)}{K_B T}\right)}$$

$$\bar{\mu} = \frac{m_J g \mu_B \exp\left(\frac{g \mu_B m_J \mu_o H}{K_B T}\right)}{\exp\left(\frac{(g \mu_B m_J \mu_o H)}{K_B T}\right)}$$

$$\bar{\mu} = \frac{g \mu_B \sum_{m_J=-J}^J \exp\left(\frac{(g \mu_B m_J \mu_o H)}{K_B T}\right)}{\sum_{m_J=-J}^J \exp\left(\frac{(g \mu_B m_J \mu_o H)}{K_B T}\right)} \quad (10)$$

The total magnetization is given by

M= number of dipoles per unit volume X average dipole moment

$$M=N \bar{\mu} \quad (11)$$

Sub eqn. 10 in 11 we get

Total magnetization as

$$M = \frac{N g \mu_B \sum_{m_J=-J}^J m_J \exp\left(\frac{g \mu_B m_J \mu_o H}{K_B T}\right)}{\sum_{m_J=-J}^J \exp\left(\frac{(g \mu_B m_J \mu_o H)}{K_B T}\right)} \quad (12)$$

Taking $x = \frac{g \mu_B \mu_o H}{K_B T}$ eqn. 12 can be written as

$$M = \frac{N g \mu_B \sum_{m_J=-J}^J m_J \exp(m_J x)}{\sum_{m_J=-J}^J \exp(m_J x)} \quad (13)$$

Eqn. (13) can be written as

$$M = N g \mu_B \frac{d}{dx} [\ln \sum_{m_J=-J}^J m_J \exp(m_J x)] \quad (14)$$

Eqn. 14 can also be written as

$$M = N g \mu_B \frac{d}{dx} [\log e^{Jx} [1 + e^{-x} + e^{-2x} \dots \dots \dots + e^{-2Jx}]] \quad (15)$$

Using the concept of geometrical progression series, the differential part of the above the can be solved by taking $y = e^{-x}$.

On solving the differential part of eqn. (15), it will reduce to

$$\frac{d}{dx} [\log e^{Jx} [1 + e^{-x} + e^{-2x} \dots \dots \dots + e^{-2Jx}]] = \frac{(2J+1)}{2} \cot \frac{h(2J+1)x}{2} - \frac{1}{2} \cot \frac{hx}{2} \quad (16)$$

And let us put $a = xJ$ or $x = a/J$

Then

$$M = N g \mu_B \left[\frac{(2J+1)}{2} \cot \frac{h(2J+1)a}{2J} - \frac{1}{2} \cot \frac{ha}{2J} \right]$$

Multiply and divide by J in the above eqn.,

$$M = N g \mu_B \left[\frac{1}{J} \cdot \left[\frac{(2J+1)}{2} \cot \frac{h(2J+1)a}{2J} - \frac{1}{2} \cot \frac{ha}{2J} \right] \right]$$

$$M = N g \mu_B J \left[\frac{(2J+1)}{2J} \cot \frac{h(2J+1)a}{2J} - \frac{1}{2J} \cot \frac{ha}{2J} \right] \quad (17)$$

We know that $x = \frac{g \mu_B \mu_0 H}{k_B T}$ also $a = x J$

Therefore, $a = \frac{J g \mu_B \mu_0 H}{k_B T}$

Eqn(17) can be written as

Therefore, $M = N g \mu_B J B_J(a)$ (18)

Where $B_J(a) = \frac{(2J+1)}{2J} \cot \frac{h(2J+1)a}{2J} - \frac{1}{2J} \cot \frac{ha}{2J}$ (19)

Considering eqn. (18) $M = N g \mu_B J B_J(a) = M_s B_J(a)$

Where, M_s is called as Saturation magnetization.

$$\text{i.e., } \frac{M}{M_s} = B_J(a)$$

Therefore, $\frac{M}{M_s} = \frac{(2J+1)}{2J} \cot \frac{h(2J+1)a}{2J} - \frac{1}{2J} \cot \frac{ha}{2J}$ (20)

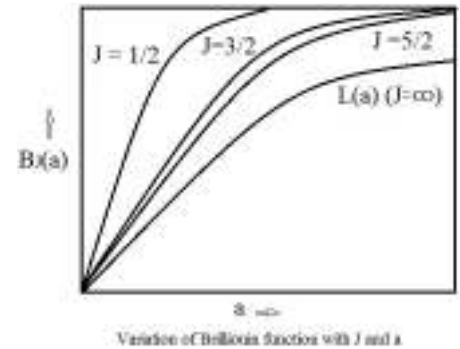
For $J = 1/2$, $\frac{M}{M_s} = \tanh a = a$ (21)

For $J = \infty$, $\frac{M}{M_s} = \left(\cot ha - \frac{1}{a} \right) = L(a)$ (22)

Where $L(a)$ is called Langevin function.

The variation of Brillouin function $B_J(a)$ with a and J is shown below

From the above discussion it is clear that if J there is infinite number of possible orientations and hence Brillouin expression becomes langevin function which is a classical explanation.



Derivation of Curie law of Paramagnetism based on quantum theory

We know that Brillouin function is given by

$$B_J(a) = \frac{(2J+1)}{2J} \coth \left(\frac{(2J+1)}{2J} a \right) - \frac{1}{2J} \coth \left(\frac{a}{2J} \right) \quad (1)$$

Equation (1) can be simplified as

$$B_J(a) = \frac{J+1}{3J} a \quad (2)$$

$$\text{We know that } M = N g \mu_B J B_J(a) \quad (3)$$

$$\text{Sub. Eqn. (2) in (3) } M = N g \mu_B J \left(\frac{J+1}{3J} \right) a \quad (4)$$

$$\text{Also we know that } a = \frac{J g \mu_B \mu_o H}{K_B T} \quad (5)$$

$$\text{Sub (5) in (4) } M = N g \mu_B J \left(\frac{J+1}{3J} \right) \frac{J g \mu_B \mu_o H}{K_B T}$$

$$\text{When cancelling J, } M = \frac{N g^2 \mu_B^2 J(J+1) \mu_o H}{3 K_B T}$$

$$\text{We know that, } \chi = \frac{M}{H} = \frac{N g^2 \mu_B^2 J(J+1) \mu_o}{3 K_B T}$$

$$\chi = \frac{M}{H} = \frac{C}{T} \quad \text{where } C = \frac{N g^2 \mu_B^2 J(J+1) \mu_o}{3 K_B} \quad (6)$$

Eqn. (6) represents Curie law according to this equation the magnetic susceptibility is inversely proportional to the absolute temperature. This equation shows that at low temperature susceptibility is high.

Quantum theory of Ferromagnetism – concept of spontaneous Magnetisation

Deduction of Curie – Weiss law of Ferromagnetic and Temperature dependence of spontaneous Magnetisation

Weiss in 1907 postulated that in ferromagnetic materials the molecular magnets in the ferromagnetic material are aligned in such a way that they exhibit a magnetization even in the absence of an external magnetic field. This is called spontaneous magnetization.

$$\text{We know that the effective applied field } H_e = H + H_i \quad (1)$$

Where H_i is called Weiss molecular field which is proportional to the magnetization M .

$$\text{Therefore, } H_i \propto M \text{ (or) } H_i = \lambda M \quad (2)$$

Where λ is called internal field constant. Sub. In eqn. (2) in (1)

$$H_e = H + \lambda M \quad (3)$$

An atom in the magnetic material has a magnetic moment of the order of one magneton, then the density is given by the quantum theory of paramagnetism is known that

$$\frac{M}{M_s} = B_J(a)$$

$$M = M_s B_J(a)$$

We know that $M_s = N g \mu_B J$

$$\text{Therefore, } M = N g \mu_B J B_J(a) \quad (4)$$

Where, $B_J(a)$ is called as Brillouin function given by

$$B_J(a) = \frac{(2J+1)}{2J} \coth\left(\frac{(2J+1)}{2J}a\right) - \frac{1}{2J} \coth\left(\frac{a}{2J}\right) \quad (5)$$

$$\text{Eqn. (5) can be simplified as } B_J(a) = \frac{J+1}{3J} a \quad (6)$$

Sub eqn. (6) in (4) we get,

$$\begin{aligned} M &= \frac{N g \mu_B J (J+1) a}{3J} \\ M &= \frac{N g \mu_B (J+1) a}{3} \end{aligned} \quad (7)$$

$$\text{We know that } a = \frac{J g \mu_B \mu_o H_e}{K_B T} \quad (8)$$

Sub. Eqn. (8) in (7) we get,

$$\begin{aligned} M &= \frac{N g \mu_B (J+1)}{3} \left[\frac{J g \mu_B \mu_o H_e}{K_B T} \right] \\ M &= \frac{N g \mu_B (J+1)}{3} \left[\frac{J g \mu_B \mu_o H_e}{K_B T} \right] \end{aligned} \quad (9)$$

Sub. Eqn. (3) in (9)

$$\begin{aligned} M &= \frac{N g \mu_B (J+1)}{3} \left[\frac{J g \mu_B \mu_o (H + \lambda M)}{K_B T} \right] \\ M &= \frac{N g^2 \mu_B^2 \mu_o (J+1) J (H + \lambda M)}{3 K_B T} \end{aligned}$$

Let $\mu_j^2 = g^2 \mu_B^2 J(J+1)$, the above eqn. becomes,

$$\begin{aligned} M &= \frac{N \mu_j^2 \mu_o (H + \lambda M)}{3 K_B T} \\ M &= \frac{N \mu_j^2 \mu_o H}{3 K_B T} + \frac{N \mu_j^2 \mu_o \lambda M}{3 K_B T} \\ M \left(1 - \frac{N \mu_j^2 \mu_o \lambda}{3 K_B T} \right) &= \frac{N \mu_j^2 \mu_o H}{3 K_B T} \end{aligned}$$

Let $\theta = \frac{N \mu_j^2 \mu_o \lambda}{3 K_B}$, then the above eqn. becomes

$$M \left(1 - \frac{\theta}{T} \right) = \frac{N \mu_j^2 \mu_o H}{3 K_B T}$$

$$M = \frac{N \mu_J^2 \mu_0 H}{3 K_B T} \frac{1}{\left(1 - \frac{\theta}{T}\right)}$$

Hence, $M = \frac{C H}{\left(1 - \frac{\theta}{T}\right)}$ where, $C = \frac{N \mu_J^2 \mu_0}{3 K_B T}$ also $C = \frac{\theta}{T}$

Therefore, $\chi = \frac{M}{H} = \frac{C}{T - \theta}$ (10)

Where C is called curie constant and θ is called ferromagnetic Curie temperature

In eqn.(10) when $T \geq \theta$ there is no spontaneous magnetization and here ferromagnetic changes to paramagnetic

When $T < \theta$ the material possess spontaneous magnetization and behave as a ferromagnetic.

To understand the above changes let us draw a graph between $\frac{M(T)}{M_S}$ Vs 'a',

For that let us consider the equation

$$M = M_S B_J(a) \quad \text{or} \quad \frac{M}{M_S} = B_J(a) \quad (11)$$

Where $B_J(a)$ is called Brillouin function equal to

$$B_J(a) = \frac{(2J+1)}{2J} \coth\left(\frac{(2J+1)}{2J} a\right) - \frac{1}{2J} \coth\left(\frac{a}{2J}\right)$$

Where, $a = \frac{J \mu_B \mu_0 H}{K_B T}$

By Weiss molecular field theory we can write 'a' as

$$a = \frac{J \mu_B \mu_0 H_e}{K_B T} \quad (12)$$

We know that $H_e = H + \lambda M$ (13)

Sub. Eqn. (12) in (13)

$$a = \frac{J \mu_B \mu_0 (H + \lambda M)}{K_B T} \quad (14)$$

We know that the spontaneous magnetization means the magnetization even in the absence of applied field i.e. $H = 0$, therefore eqn. (14) becomes

$$a = \frac{J \mu_B \mu_0 \lambda M(T)}{K_B T} \quad (15)$$

In eqn. (15) we write $M(T)$ in place of M just to remind that M is temperature dependent.

We know that saturation magnetization

$$M_S = g \mu_B J N \quad (16)$$

From eqn. (15) we can write

$$M(T) = \frac{a K_B T}{g \mu_B \mu_0 J \lambda} \quad (17)$$

Dividing eqn. (17) by (16) we get

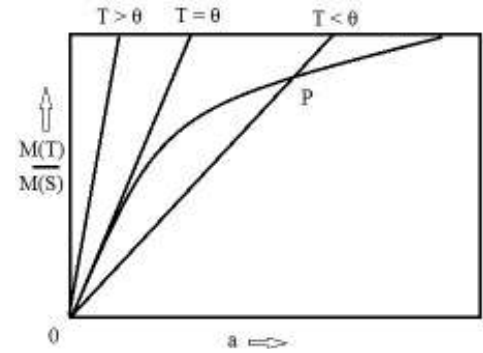
$$\begin{aligned} \frac{M(T)}{M(S)} &= \frac{\frac{a K_B T}{g \mu_B \mu_0 J \lambda}}{\frac{g \mu_B J \lambda N}{g \mu_B J \lambda N}} \\ \frac{M(T)}{M(S)} &= \frac{a K_B T}{g \mu_B \mu_0 J \lambda} \times \frac{1}{g \mu_B J \lambda} \\ \frac{M(T)}{M(S)} &= \frac{a K_B T}{g^2 \mu_B^2 \mu_0 J^2 \lambda N} \end{aligned} \quad (18)$$

Eqn. (11) can be written as

$$\frac{M(T)}{M(S)} = B_J(a) \quad (19)$$

The graphical solution of simultaneous equations (18) (19) is depicted in the below figure

It should be noted that for $a \rightarrow \infty$ (which corresponds to $T \rightarrow 0$, $B(a) \rightarrow 1$). Thus thick curve in the figure represents eqn.(19). Eqn. 18 represents a straight line passing through the origin with a slope proportional to T .



The straight lines in the figure represent eqn. (18) for $T < \theta$, $T = \theta$ and $T > \theta$, where θ is a critical temperature, all with $\mu_0 = 0$. The solution of the two equations is of course given by the point of intersection of the two curves.

For $T < \theta$, a non vanishing value of M is obtained corresponding to point P even though the external field H is zero. This is nothing but the spontaneous magnetization. It should be noted that possible solution corresponding the point '0' leads to unstable conditions and here is rejected.

For $T = \theta$, the slope of straight line is equal to that of the tangent to the thick curve and thus no solution is possible. It means θ is the critical temperature. This θ is actually called ferromagnetic Curie temperature.

For $T > \theta$, the two curves do not intersect and thus there is no solution possible and hence no spontaneous magnetizations can occur.

Temperature dependence of spontaneous Magnetization

Considering eqn. (18)

$$\frac{M(T)}{M(0)} = \frac{a K_B T}{g^2 \mu_B^2 \mu_0 J^2 \lambda N} \quad (1)$$

We know that Internal field constant $\lambda = \frac{3 K_B \theta_f}{N \mu_0 \mu_J^2 J^2}$ or $\frac{1}{\lambda} = \frac{N \mu_0 \mu_J^2}{3 K_B \theta_f}$ (2)

Substitute eqn. (2) in eqn. (1)

$$\begin{aligned} \frac{M(T)}{M(0)} &= \frac{a K_B T}{g^2 \mu_B^2 \mu_0 J^2 N} \times \frac{N \mu_0 \mu_J^2}{3 K_B \theta_f} \\ \frac{M(T)}{M(0)} &= \frac{a T \mu_J^2}{3 g^2 \mu_B^2 J^2 K_B \theta_f} \end{aligned} \quad (3)$$

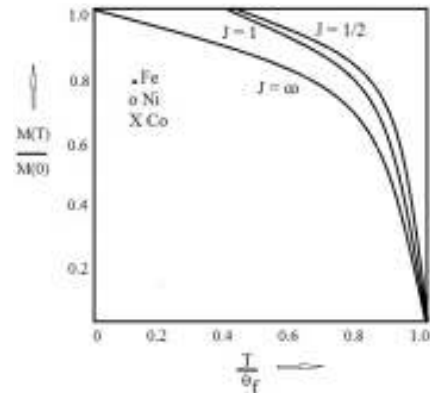
We know that $\mu_J^2 = g^2 \mu_B^2 J(J+1)$ (4)

Sub eqn. (4) in (3) we get

$$\begin{aligned} \frac{M(T)}{M(0)} &= \frac{a T g^2 \mu_B^2 J(J+1)}{3 g^2 \mu_B^2 J^2 K_B \theta_f} \\ \frac{M(T)}{M(0)} &= \frac{a T (J+1)}{3 J \theta_f} \end{aligned} \quad (5a)$$

The quantity $\frac{M(T)}{M(0)}$ must satisfy both Equations (5a) and (12)

It is important to note that for a given value of J, this procedure leads to a universal curve when $\frac{M(T)}{M(0)}$ is plotted as a function of $\frac{T}{\theta_f}$. In the above graph curves for $J=1/2$, $J=1$ and $J=\infty$ are represented the latter case corresponds to classify freely rotating dipoles. It is observed that the curve for $J=1/2$ fits the data best, indicating that the magnetization is essentially with the electron spins rather with orbital momentum of electrons.

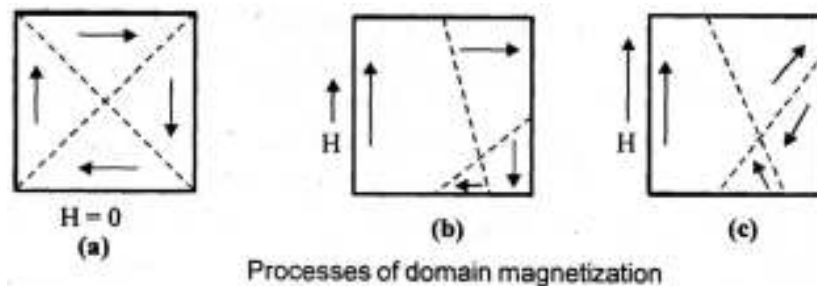


Domain Theory of Ferromagnetism

The explanation of the hysteresis property of the ferromagnetic materials is based on domain concept proposed by Weiss. This concept is called domain theory of Ferromagnetism.

According to him a magnetic domain is a region of a crystal in which all the spin magnetic moments are aligned to produce magnetic moments in only one direction. A small localized region in a ferromagnetic material is also known as domain. The size of a domain varies from to the entire size of the crystal. A domain acts as a single magnetic dipole. The dipole moments of each and every domain are painted in random direction, and the net magnetization, in the absence of the field is zero (Fig.a).

These domains are separated from other domains by a well known as domain wall or Bloch wall. When a magnetic field is applied to a magnetic material, the domains that are parallel to the applied field increase in size at the expense of the other domains (Fig.b). The size of the other domains which are not parallel to the applied field decreases. As shown in the below figure.



From the above figure we conclude that when the magnetic field is applied to a ferromagnetic material the magnetization occurs in the specimen by two ways

1. By the movement of domain walls
2. By the rotation of domain walls (Fig.c).

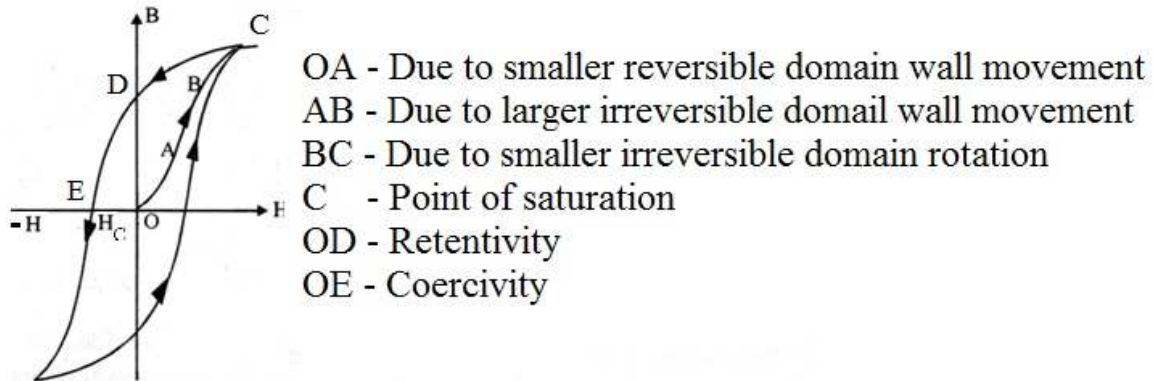
Explanation of Ferromagnetic Hysteresis on the basis of Domains

We know when the ferromagnetic material is subjected to external field, there is an increase in the value of the resultant magnetic moment due to two process

- (i) The movement of domain walls

(ii) Rotation of domain walls

When a small external field is applied the domains walls is displaced slightly in the easy direction of magnetization. This gives rise to small magnetization corresponding to the initial portion of the hysteresis curve OA as shown in the below figure



Now, if the applied field is removed, then the domains return to its original state and is known as reversible domains. When the field is increased, large number of domains contributes to the magnetization and thus the magnetization increases rapidly with H.

Now, even when the field is removed, because of the displacement of domain wall to a very large distance, the domain boundaries do not come back to their original position. This process is indicated as AB in the above figure and this domain is called irreversible domains.

At point 'B' all the domains have got magnetized along the easy direction.

Now, when the field is increased, the domains start rotating along the field direction and the anisotropic energy is stored in the hard direction represented as BC as shown in the figure

Thus the specimen is said to attain the maximum magnetization. At this position, even after the removal of external field the material posses maximum magnetization, called residual magnetism (or) represented, represented by OD in the above figure

Actually after the removal of the external field, the specimen will try to attain the original configuration by the movement of Bloch wall. But this movement is stopped due to the presence of impurities, lattice imperfections etc. Therefore to overcome this, a

large amount of reverse magnetic field is applied to the specimen. The amount of energy spent to reduce the magnetization to zero is called as coercivity represented by 'OE' in the figure

Hysteresis:

When a ferromagnetic material is made to undergo through a cycle of magnetization, the variation of B (magnetic induction) with respect to H (Applied field) can be represented by a closed hysteresis loop (or) curve i.e. it refers to the lagging of magnetization behind the magnetizing field.

Hysteresis Loss:

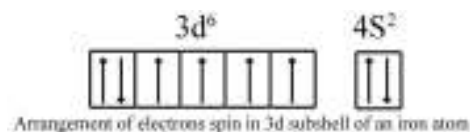
It is the loss of energy in taking a ferromagnetic specimen through a complete cycle of magnetization and the area in closed is called hysteresis loop.

Based on the area of the hysteric loop the magnetic materials are classified into soft and hard magnetic materials

Heisenberg's theory of Exchange Interaction

The magnetization in a ferromagnetic material is produced due to the existence of magnetic spin moments and not due to the magnetic orbital moment.

Consider an iron atom. Its electronic configuration is $1S^2, 2S^2, 2P^6, 3P^6, 3d^6, 4S^2$ i.e, in iron, the 3d sub shell is an unfilled one as shown below.



There are five orbital in the sub shell of these five orbital, there are four unpaired electrons. These electrons are aligned parallel in separate orbitals due to the Pauli's exclusion principle and electrostatic energy and not because of the magnetic interaction. In an iron atom, the four unpaired electrons in 3d orbital take different m_s values but the same value. The electrostatic interaction energy and Paul's exclusion principle are put together to constitute a new kind of interaction known as exchange interaction.

The large value of Weiss internal field in a ferromagnetic material can be explained with the help of exchange interaction. Exchange interaction is a quantum mechanical concept.

The exchange interaction between any two electrons present in different quantum states reduces the energy when they are having parallel spin. The reduction in energy is due to the consequence of Pauli's exclusion principle and Coulombic force.

The exchange interaction energy E_{ex} between any two electrons is represented using quantum mechanics as

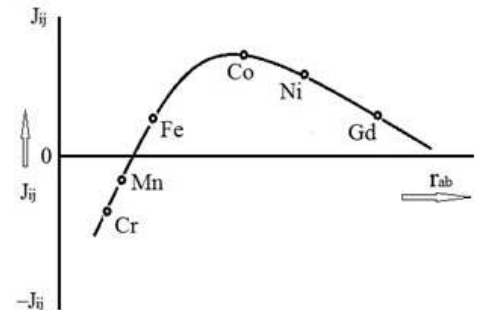
$$E_{ex} = -2 J_{ij} S_i S_j$$

Where J_{ij} is the exchange integral for the two atoms and S_i and S_j are the spin angular moments associated with i^{th} and j^{th} state.

When a plot is drawn between exchange integral value and the ratio of inter atomic distance ' r ' to the ' r_d ' is orbital radius as shown in the figure.

From the graph we can say

- 1) The Value of J_{ij} is positive when $r/r_d > 3$ i.e., the exchange energy is negative and hence the parallel orientation is high. Due to this the atom possesses ferromagnetic properties. (Examples: Fe, Co, Ni and Gd).
- 2) The value of J_{ij} is negative when $r/r_d < 3$. i.e., the exchange energy is positive and hence the atoms coming under this criterion possess anti-ferromagnetic properties (Examples: Cr, Mn).



The ratio r/r_d is the criteria for a magnetic materials whether it is Ferro (or) antiferromagnetic materials. Moreover alloys of any two materials give rise to favorable ratio value. This is a great boom in solid state for the production of new materials of any desired characteristics.

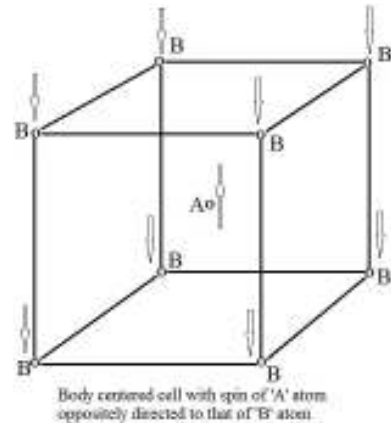
Anti – Ferromagnetic ordering:

L. Neel in 1936 state that an ordered state of crystals is possible with exchange integral is positive is exchange energy is negative where neighboring dipoles are aligned in an anti parallel arrangement. This phenomenon is named as ant ferromagnetism or ant ferromagnetic ordering.

In 1938, Bizette and others discovered using neutron diffraction method that Mno is actually a crystal which has B.C.C structure and behaves as an anti ferromagnetic substance. Other examples found, latter to behave as anti ferromagnetic are MnF_2 , FeF_2 , CoF_2 , FeO_2 , $\text{CuCl} \cdot 2\text{H}_2\text{O}$, $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$,...etc.

An example of this behavior is shown aside.

Consider a above crystal containing two types of atoms A and B distributed over two inter locking lattices. For example let the B atoms occupy the corner points of an elementary cube and A atoms being located at the centers of these cubes. Further let the interaction between the atoms be such that the A spins tend to line up anti parallel to the B spins.



If the moments at lattice points are summed over an entire crystal the result is zero. It is because one set of dipoles is equal and oppositely polarized to another set of dipoles. Therefore, in the absence of applied magnetic field, the magnetization will be zero. When field is applied, we find a small magnetization in the direction of applied field. This magnetization increases with increase in temperature up to a critical value. Beyond this critical temperature, the dipoles are randomly oriented and the substance becomes paramagnetic, obeying the curie- Weiss law with Curie temperature θ of opposite sign to that found in ferromagnetism.

“The critical temperature below which an anti ferromagnetic substance shows a spontaneous magnetization and beyond which it reduces to paramagnetic is called “Neel temperature” (T_N).

Structure and Applications of Ferrites

Ferrites are the magnetic compounds consisting of two or more different kinds of atoms. Generally ferrites are expressed as $X^{2+} Fe_2^{3+} O_4$ where X^{2+} stands for suitable divalent metal ion such as Mg^{2+} , Zn^{2+} , Fe^{2+} , Mn^{2+} , Ni^{2+} , etc.

Examples

(i) If X^{2+} is replaced by Ni^{2+} then the ferrite ($Ni^{2+} Fe_2^{3+} O_4$) is formed, thus named as nickel Ferrite.

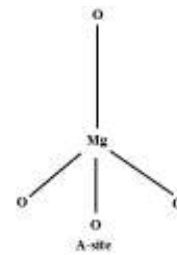
(ii) If X^{2+} is replaced by Fe^{2+} then the Ferrite ($Fe^{2+} Fe_2^{3+} O_4$) is formed, thus named as ferrous ferrite.

Ferrites formed usually have a face centered cubic structure of oxygen ions closely packed together with the divalent and trivalent metal ions in the interstitial sites. This structure is called spinal structure. There are two types of ferrite structures

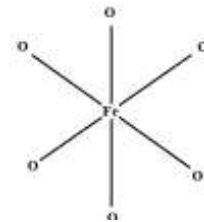
- (i) Regular spinal and
- (ii) Inverse spinal

Regular spinal: Eg. $Mg^{2+} Fe^{3+} O_4$

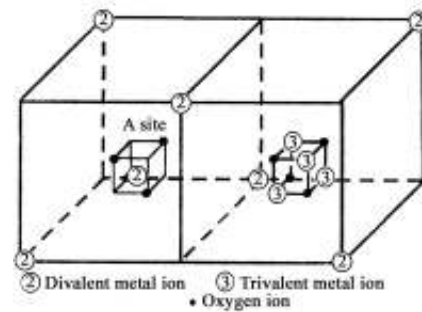
In this type each metal atom (divalent) is surrounded by four O^{2-} ions in a tetragonal fashion. For example, if the metal atom is Mg^{2+} , then the structure is as shown in the figure and it is called 'A' site. Totally in a unit cell there will be 8 tetrahedral (8A) sites.



Each Fe^{3+} (trivalent) is surrounded by six O^{2-} ions and forms an octahedral fashion as shown below. Totally, there will be 16 such octahedral sites in the unit cell. This is located by 'B' site.

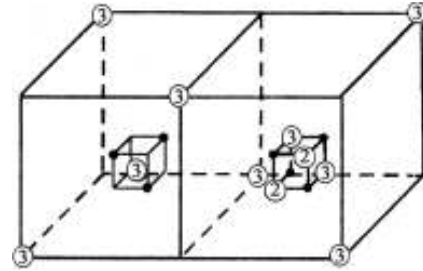


Thus in a regular spinal, each divalent metal ion (Mg^{2+}) exists in a tetrahedral form (A site) and each trivalent metal ion (Fe^{3+}) exists in an octahedral form (B site). Hence the sites A and B combine together to form a regular spinal ferrite structures as shown below.



Inverse spinel: Example. $\text{Fe}^{3+} [\text{Fe}^{2+} \text{Fe}^{3+}] \text{O}^4$

In this the Fe^{3+} ions (trivalent) occupies all the A sites (tetrahedral) and half of the B sites (octahedral) also. Thus the left out B sites will be occupied by the divalent (Fe^{2+}). The inverse spinel structure is shown below.



Types of interaction present in the ferrites.

The spin arrangement between the A site and B site is in an anti parallel manner and it was explained by Neel. According to him, in ferrites, the spin arrangement is anti parallel and there exists some interaction between the A and B sites which is represented as AB interaction.

Apart from this, there are two more interactions (ie) AA and BB interaction which is negative and considerably weaker than AB interactions.

The tendency of AB interaction is to align all A spins parallel to each other and anti parallel to all B spins. But the tendency of AA and BB interaction is to spoil the parallel arrangement of A and B spins respectively.

Since AB is Very strong as compared with AA and BB, the effect of AB interaction dominates and gives rise to anti parallel spin arrangement.

Applications of Ferrites:

1. They are used to produce Ultrasonic by magneto strict ion principle.
2. Ferrites are used in audio and video transformers (Ni – Zn ferrites)
3. Ferrite rods are used in radio receivers to increase the sensitivity
4. Since the ferrites have low hysteresis loss and eddy current loss, they are used in two port device such as gyrator, circulator and isolator.
5. They are also used for power limiting and harmonic generation.
6. Ferrites are used in parametric amplifiers so that the input can be amplified with low noise figures.
7. They are used in computers and data processing circuits.

8. Ferro cubes (ferrites with rectangular hysteric loop) are used in switching circuits and in matrix storage devices of computers.
9. Ferrites are not metals, but their resistivities lies in the range of insulator or semiconductor. Thus, the power loss due to eddy currents is reduced in this type of materials and hence they are used in microwave frequency applications.

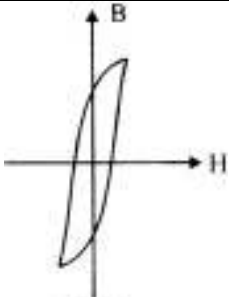
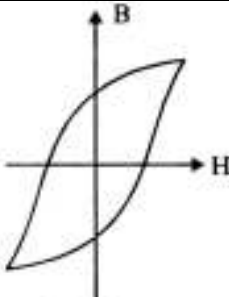
Soft and Hard Magnetic Materials. Soft magnetic materials:

The materials which can be easily magnetized and demagnetized are called soft magnetic materials.

Hard magnetic materials:

The materials which are very difficult to magnetic and demagnetize are called hard magnetic materials.

Differences between Soft and Hard

Sl.No.	Soft	Hard
1.	They can be easily Magnetized and demagnetized	They cannot be easily Magnetized and demagnetized
2.	Movement of domain wall is easy and hence even for a small applied field large Magnetization occurs	Movement of domain wall is not easy due to the presence of impurities and hence large field is required for magnetization
3.	The hysteric loop is very narrow as shown below	The hysteric loop is very broad as shown below
	 Soft iron	 Steel

4.	Loop area is less and hence hysteresis loss is minimum	The loop area is large and hence the hysteresis loss is maximum
5.	Susceptibility and permeability are high	Susceptibility and permeability are low.
6.	Retentivity and coercivity are small	Retentivity and coercivity are large.
7.	They have low eddy current loss.	They have high eddy current loss.
8.	These materials are free from irregularities like strain or impurities	These materials have large amount of impurities and lattice defects.
9.	Examples: Iron, Silicon alloys, Ferrites, Garnets etc.	Carbon Steel, Tungsten steel Cu-Ni-Fe (cunife) Cu-Ni-Co(cunico) Al-Ni-Co(Alnico)
10.	Applications: It is used to make the temporary magnets. They are also used in switching devices, electro magnets storage of computers	Applications: It is used to make the permanent magnets which are used in magnetic detectors, microphones and magnetic separators.

Magnetic Storage Devices:

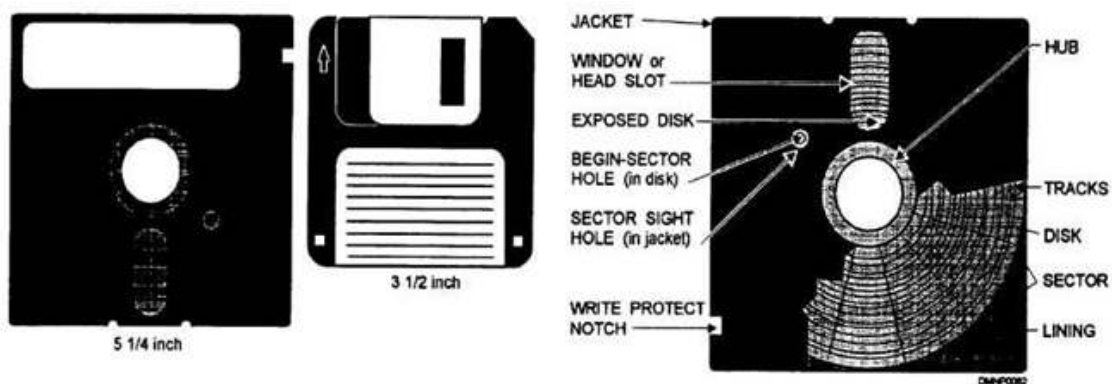
Floppy Disk:

The hard disks are suitable only for large and medium sized computers and often are too expensive for small computer systems. Floppy disks are used in small computer systems as secondary storage devices.

It is made up of flexible plastic material and hence called as Floppy disk. It is also called as Diskette. It acts both as an input and output device. A floppy disk is made of flexible plastic which is coated with magnetic oxide. The floppy disk is provided with a central hole. This hole is used for mounting the disk in the floppy drive

unit. The disk is permanently enclosed in a flexible square envelope. This cover prevents the disk from dust and moisture.

There is an opening called Read/write head aperture slot. The reading and writing operation is done in floppy disk through this opening. There is a small index hole in the cover and there will be a hole in the drive disk. When these two holes match then only the storage operation can be started. The write protect notch is used to prevent writing on the disk by other users which can be done by covering the notch with a sticker. Floppy of 3.5", 5.25" are shown below.



Operation:

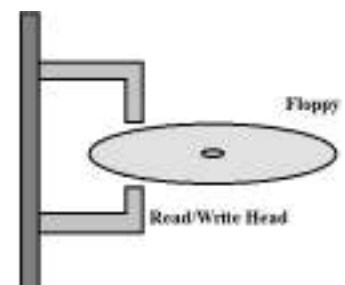
When the floppy is put into the drive and when the drive is operated, the disk inside the cover is rotated. As the disk rotates, computer data are organized into a number of concentric circles (or) track, each of which has a designated location number. There are 200 to 800 tracks on the disk surface as shown below.

The read/write operation is done by the read/write head in the drive unit. The read/write head is a small electro magnet with minute gap between the poles. Read/write head has two coils viz. Read and write coils.

Writing operation:

The floppy disk moves over the gaps as shown aside.

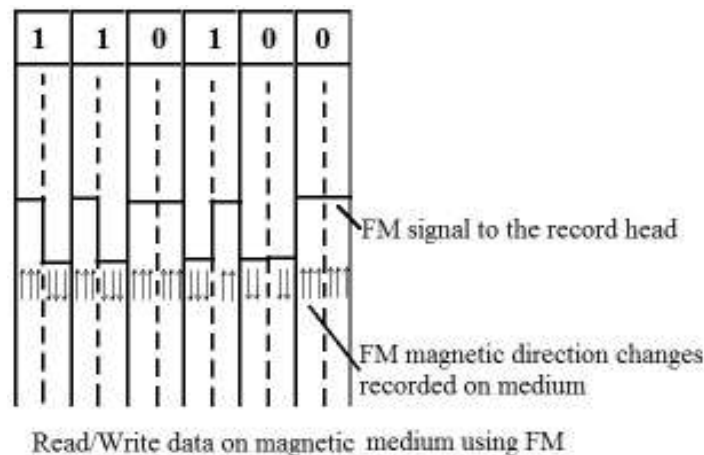
Now the electric pulses from the CPU flow through the write coil of the head and magnetize the iron oxide coating in the disk to the proper pattern.



Reading operation:

When the data is to be read the magnetized patterns induce pulses of current in the read coil and is amplified then fed to the CPU. Thus data can be stored and accessed from the floppy disk on both sides (or) single side.

Floppy disk has three different formats for writing or reading the data from the disk, namely, single density, double density and very high density. A frequency modulation (FM) is used to write/read data in single density format while modified frequency modulation (MFM) is used in the case of double density format. Write/read in the case of single density for using FM is shown below.



CD- ROM (Compact Disk -Read only Memory)

As the name itself implies that it is a type of a CD in which one can only read the data and cannot rewrite it. Hence this comes under the category of a permanent storage device used in computers.

We are all familiar about a Floppy disk, which is a magnetic storage device in which the data is stored and retrieved using magnetic pulses. Since it can store very less number of data, nowadays CD's are used.

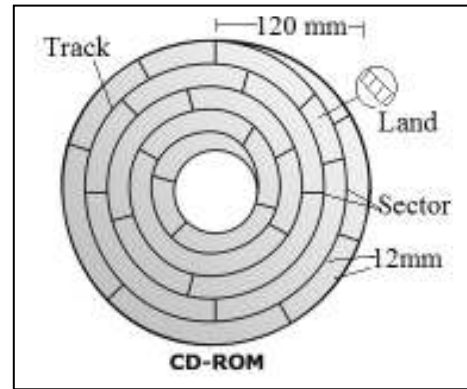
This CD – Rom is an optical storage device which is capable of storing large number of data than a floppy disk. For example one CD can store 1000's of pages from a book (or) many software also.

Principle:

CD – Rom used the principle of writing the datas using a high power laser beam and reading the datas using a low power laser beam. By forming pits (O's) and lands (1's) over the CD.

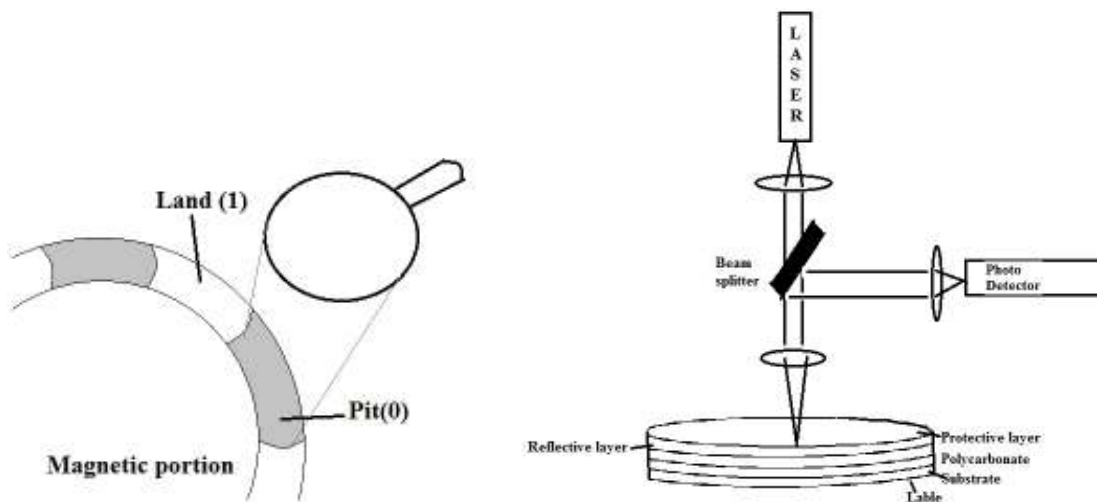
Design of a CD – Rom:

CD – Rom is a plastic (or) photo – polymer disk made up of poly – carbonate grown over the substrate. It has a diameter of 120 mm and of thickness 1-2 mm with a circular hole of diameter 15mm as shown below.



In this disk the datas can be stored in tracks and sectors, from 46mm to 116 mm of diameter. The magnified portion of data storage is indicated in the below figure.

Since we are going to use laser light for writing and reading the data, the reflectivity at the top surface should be increased. Therefore a layer of silver or aluminum is deposited as vapor over the poly – carbonate surface as shown below



A protective layer protects the CD-ROM from dust and moisture.

Recording/writing process.

The above figure shows the system used for recording/writing the datas over the CD – Rom and also for retrieving/reading the datas from it.

Initially all over the CD – Rom we shall have lands (1's) of highly reflectivity. Now to change it as 0's and 1's a high power laser is directed towards the surface of the CD – Rom through the condensing lens beam splitter and condensing lens.

This high power laser beam burns the surface of the CD – Rom and creates a small hole called pit, corresponding to 0's of 0.8 μm diameter.

By similar method we want 0's, laser should be focused and pits are created and the remaining area will be as such as lands i.e, 1's.

Therefore the pits (0's) represents the burnt area and the land (1's) represents the unburnt area in CD – Rom. Thus the datas are digitally recorded/written over the CD – Rom.

Magneto – optical Retarding:

Introduction:

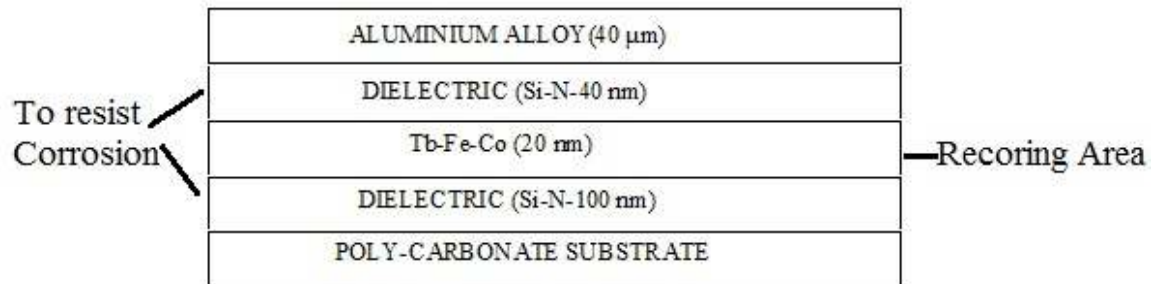
Magneto optical (Mo) recording was first reported in 1957 by Williams using MnBi film as the recording material. After various advances in tracking and focusing servo systems, laser diodes, and recording materials, a magneto – optical recording system was announced by Imamura in 1980. The first commercialization of an Mo disc was in 1988 for data recording with a capacity of 650 m byte per two – sided 5-25 inch disc.

Magneto – optical disc structure:

The most popular material for magneto optical disc retarding is Tb – Fe – Co an amorphous alloy consisting of the rare earth Tb and the transition metal alloy Fe – Co. An example of a typical composition is $\text{Tb}_{20} \text{Fe}_{74} \text{Co}_6$. In most cases, the material is doped with a small amount of for protection from corrosion.

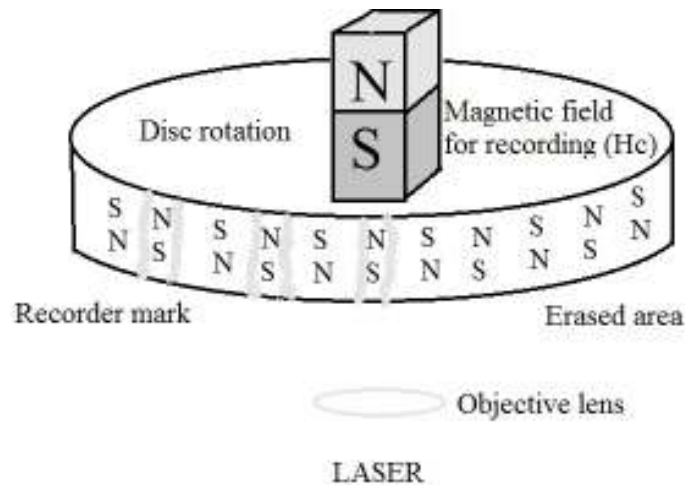
When dielectric layers of Si-N, example: Si_3N_4 are inserted on both sides of Tb – Fe – Co,

the structure is very resistant to corrosion. A Tb – Fe – Co film, dielectric layers and reflective Aluminum alloy film are deposited on a poly carbonate substrate by sputtering in Ar gas. A typical four layer structure used as the best balance between recording and readout characteristics in conventional magneto optical disc is shown below.



Recording:

Recording on an magneto optical disc is called thermo magnetic recording because the magnetization is reversed at an area heated by laser – pulse irradiation where the coercivity at the raised temperature is less than the external magnetic field. The recorded mark is a magnetic domain whose magnetization is directed upward i.e., south to North opposite to the erased state as shown in the below figure



The advantages of Tb – Fe – Co are as follows

1. Tb has a large Magnetostriction coefficient and a large single – ion anisotropy.
2. Tb has spin opposite to the spin of the transitional metal Fe – Co since Tb – Fe – Co is a ferromagnetic material.

3. The material is amorphous and therefore there are no grain boundaries to contribute to the noise.

Read out:

The signal from the recorded mark is read out using the magneto – optical Kerr effect originating in the magnetic layer.

The magneto – optical Kerr effect (MOKE) is defined as “the phenomenon that the light reflected from a magnetized material has a slightly rotated plane of polarization is called as magneto – optical Kerr effect.”

A linearly polarized light incident on the magnetic layer is used for readout. The reflected light from the magnetic layer is also linearly polarized light, but the axis of polarization is rotated at an angle called the Kerr rotation, with a typical value of the direction of the Kerr rotation for the recorded area is the opposite of that for the erased area. Thus difference is detected for signal read out in an optical system using a polarizing beam splitter.